Journal of Adolescent Health 54 (2014) 144–150

ELSEVIER

JOURNAL OF ADOLESCENT HEALTH

www.jahonline.org

Original article

Worldwide Never-Smoking Youth Susceptibility to Smoking

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Article history: Received February 13, 2013; Accepted July 30, 2013
Keywords: Tobacco use; Susceptibility to smoking; Global youth tobacco survey; Tobacco industry

See Related Editorial p. 119

A B S T R A C T

Purpose: To estimate susceptibility to smoking among never-smoking youth globally and identify factors associated with such behavior.
Methods: Cross-sectional data for 168 countries were obtained from the Global Youth Tobacco Survey. Simple and multiple logistic regression analyses were conducted. Frequencies and proportions for descriptive statistics, and adjusted odds ratios with 95% confidence intervals for logistic regression models were reported.
Results: Approximately 12.5% of never-smoking youth worldwide were susceptible to smoking worldwide, of which 7.2% were males and 5.3% were females. Compared with youth in the Americas, those in other WHO regions were associated with decreased susceptibility to smoking. Regardless of gender, exposure to parental or peer smoking, secondhand smoke inside or outside home, and tobacco industry promotion was associated with increased smoking susceptibility. In contrast, support for smoke-free policies and school antismoking education was associated with decreased susceptibility to smoking among females. Moreover, exposure to antismoking media messages was associated with increased susceptibility to smoking among never-smoking youth.
Conclusions: Approximately 1 in 8 never-smoking youth worldwide was found to be susceptible to smoking. A comprehensive approach involving parental and peer education, smoke-free policies, ban on tobacco advertising and promotions, and antismoking education in schools should be developed by policy makers and public health professionals to protect never-smoking youth from being susceptible to smoking and transforming into future regular smokers.
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IMPLICATIONS AND CONTRIBUTION

Limited research has been conducted to understand the pre-experimentation stages of cigarette learning behavior in low-and-middle-income countries. This study expands across 168 countries to estimate and identify key factors for smoking susceptibility among never-smoking adolescents worldwide. The findings have potential to inform initiatives for early tobacco intervention efforts targeted at never-smoking youth.

Tobacco use continues to be the leading preventable cause of noncommunicable diseases and deaths worldwide [1], with up to 90% of smokers beginning to smoke as adolescents [2]. If current trends persist, it is predicted that annual tobacco-related mortality worldwide will be over eight million deaths by 2030 and more than one billion cumulative deaths by the end of the 21st century, with 70% of these deaths occurring in low-and middle-income countries (LMICs) where 80% of current smokers live [3]. Therefore, along with strong tobacco control strategies and successful prevention efforts targeting smokers, it is important to prevent never-smoking youth from being susceptible to smoking and initiating smoking behavior; however, only a few studies have

Competing Interests: The authors have no competing interests to declare.
Funding: The authors declare that no external funds were used to conduct this study.
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http://dx.doi.org/10.1016/j.jadohealth.2013.07.036

examined never-smoking youth susceptibility to smoking in LMICs [4–6].

Cigarette smoking, the dominant form of tobacco use worldwide, is a learned behavior and progresses through several stages—preparation, initiation, experimentation, regular smoking, and addiction [7]. The preparation stage refers to a period during which a person starts thinking about smoking and establishes ideas and beliefs about cigarette use, and the initiation stage refers to a period when a person smoked an entire cigarette. Although limited research had been done, the concept of the preparation and initiation stages did not receive significant recognition until the 1994 U.S. Surgeon General report on youth smoking [8]. Since then, research in this area has continued to grow but still receives less attention in LMICs [4–6,9]. Susceptibility to smoking is defined as lack of firm decision against smoking and usually starts in the preparation and/or initiation stages of smoking behavior [10]. Earlier studies identified susceptibility to smoking as a stronger predictor of smoking experimentation than familial smoking behavior [9–12], with adolescents more likely to become regular or established smokers [12,13], with increased participation in tobacco industry promotion campaigns [9,14], but decreased responsiveness and compliance with tobacco prevention programs [15]. Therefore, an in-depth understanding of the preparation and initiation stages of smoking has strong potential to provide directions for tobacco control strategies and prevention efforts to prevent never-smoking youth from initiating their first cigarette. Thus, this study aims to estimate the susceptibility to smoking among never-smoking youth worldwide and identify key factors associated with such behavior.

Methods

We utilized data from the Global Youth Tobacco Survey (GYTS), 1999–2008, to conduct this study. The evolution, design, and methodology of the GYTS had been described in detail in earlier studies [16]. In brief, the GYTS is a school-based survey and employs a two-stage cluster sampling design to collect representative data on tobacco use among youth aged 13–15 years. The clusters of schools proportional to student enrollment were selected during the first stage, followed by random selection of classes during the second stage. The survey questionnaire was administered to all students in selected classes. Study participation was voluntary and anonymous using self-administered data collecting field procedures. The school, class, and student response rates were calculated for each country. For this study, we collected information on youth tobacco use from 168 countries (see online Supplementary Appendix A) surveyed during 1999–2008 from the publicly available Centers for Disease Control and Prevention - Global Tobacco Surveillance System (CDC - GTSS) Data portal [17]. Consistent with earlier studies, we used the most recent survey for countries where the survey has been conducted more than once [18,19].

The dependent variable is susceptibility to smoking among never-smoking youth and was obtained using the algorithm developed by Pierce et al. [10,20]. Never-smoking youth were identified as those participants who responded "no" when asked whether they had ever tried or experimented with cigarette smoking, even one or two puffs. We defined susceptibility to smoking by participants' responses to the following four-point ordinal scale: (1) definitely not; (2) probably not; (3) probably yes; and (4) definitely yes, as based on the following three questions:

1) "If one of your best friends offered you a cigarette, would you smoke it?"
2) "Do you think you will be smoking cigarettes 5 years from now?"
3) "At any time during the next 12 months do you think you will smoke a cigarette?"

Using previously described methods [18], those who responded "definitely not" to all three questions were considered as nonsusceptible and recoded as '0', and all other adolescents were considered susceptible to smoking and recoded as '1' (Table 1).

Based on the existing literature [21,22], we constructed the following 11 independent variables that could potentially be associated with susceptibility to smoking among never-smoking youth: (1) parental or peer smoking; (2) exposure to secondhand smoke (SHS) inside the home; (3) exposure to SHS outside the home; (4) support for smoke-free policies (SFPs); (5) knowledge about the harmful effects of smoking and SHS; (6) exposure to antismoking media messages; (7) tobacco industry promotions; (8) school antismoking education; (9) age; (10) sex; and (11) World Health Organization (WHO) region. The "WHO region" was included to identify differences in susceptibility to smoking among never-smoking youth across the six WHO regions: Africa, East Mediterranean, Europe, The Americas, Southeast Asia, and Western Pacific (see online Supplementary Appendix A). All variables were categorical in nature, and most responses were dichotomized using previously described methods [18,23] (Table 1). To estimate susceptibility to smoking among never-smoking youth worldwide, the data were weighted to adjust for design effects, nonresponses, and poststratification of the sample relative to the grade and sex distribution in the population. A final adjustment summed the weights by grade and gender to the population of school-going adolescents in each sample site. The methodology used for weighing the measures has been previously used [18,24] and described by the CDC in the help file of the GYTS [25].

Analyses were conducted using SAS 9.2 statistical software with "survey" syntaxes to accommodate for sampling weights (SAS Institute Inc. Cary, NC). Descriptive statistics were reported using frequencies and proportions. Bivariate analyses using a *t*-test for continuous variables and a χ^2 test for categorical variables were conducted to determine the relationship between the dependent and independent variables. Finally, a multivariable logistic regression model was conducted to identify factors associated with susceptibility to smoking among never-smoking youth. Adjusted odds ratios (AORs) and 95% confidence intervals (CI) were reported. Based on the established gender-differences in smoking behavior and smoking susceptibility among adolescents, analyses were conducted separately for males and females, respectively [24,26]. Similar to the total study population analysis, results for bivariate and multivariable logistic regression models were reported. A two-sided 5% significance level was used for all statistical inferences. The study was approved by the institutional review board at East Tennessee State University.

Results

The study sample included 356,414 never-smoking school-going adolescents aged 13–15 years from 168 countries, representing a total of 70,025,268 never-smoking youth worldwide. Overall, 12.5% of them were susceptible to smoking, of which 7.2%

Table 1
Study measures, survey items with responses, Global Youth Tobacco Survey (GYTS)

Study measure		GYTS survey items	GYTS item responses	Dichotomized measure
GYTS question used to obtain never-smoking youth				
Smoking status		Have you ever tried or experimented with cigarette smoking, even one or two puffs?	Yes No	Yes = Ever smoker No = Never smoker
Dependent variable				
Smoking susceptibility		If one of your friends offered you a cigarette, would you smoke it? At any time during the next 12 months do you think you will smoke a cigarette? Do you think you will be smoking cigarettes 5 years from now?	Definitely not Probably not Probably yes Definitely yes	No = "Definitely not" for all three items Yes = any other responses for any of the three items
Independent variables				
Exposure to secondhand smoke (SHS) inside home		During the past week, on how many days have people smoked in your home, in your presence?	0 1 to 2 3 to 4 5 to 6 7	No = 0 days for both items Yes ≥1 day for both items
Exposure to SHS outside home		During the past 7 days, on how many days have people smoked in your presence, in places other than in your home?	0 1 to 2 3 to 4 5 to 6 7	No = 0 days for both items Yes ≥1 day for both items
Parental or peer smoking	Parents Peers	Do your parents smoke? Do any of your closest friends smoke cigarettes?	None Both Father only Mother only None of them Some of them Most of them All of them	No = "none" or "none of them" for both items Yes = any other reasons for either item
Support for smoke-free policy (SFP)		Are you in favor of banning smoking in public places (such as in restaurants, in buses, streetcars, and trains, in schools, on playgrounds, in gyms and sports arenas, in discos)?	No Yes	No Yes
Knowledge about harmful effects of smoking and SHS		Do you think cigarette smoking is harmful to your health? Do you think the smoke from other people's cigarettes is harmful to you?	Definitely not Probably not Probably yes Definitely yes	No = "Definitely not" for both items Yes = any other responses for either item
Exposure to antismoking media messages		During the past 30 days (1 month), how many antismoking media messages (e.g., television, radio, billboards, posters, newspapers, magazines, movies) have you seen? When you go to sports events, fairs, concerts, community events, or social gatherings, how often do you see antismoking messages?	None A few A lot	No = "none/never" response for both items Yes = any other response for either item
Tobacco industry promotion		Do you have something (t-shirt, pen, backpack, etc.) with a cigarette brand logo on it? Has a (cigarette representative) ever offered you a free cigarette?	No Yes	No Yes
Antismoking school education		During this school year, were you taught in any of your classes about the dangers of smoking? During this school year, did you discuss in any of your classes the reasons why people your age smoke? During this school year, were you taught in any of your classes about the effects of smoking, such asit makes your teeth yellow, causes wrinkles, or makes you smell bad?	No Not sure Yes	No = "No" for all three items Yes = Yes for any of the three items
Age		How old are you?	13 years 14 years 15 years	
Sex		What is your sex?	Female Male	No = Female Yes = Male

Dichotomization of most GYTS survey responses conducted using previously described methods [20,21].
"No" responses were coded as "0" and "Yes" responses as "1" for multivariable logistic regression model.

were males and 5.3% were females, respectively. Susceptibility to smoking among never-smoking youth varied significantly across the six WHO regions, ranging from 10.1% in Southeast Asia to 29.8% in Europe ($p < .0001$) (Table 2). Approximately 27% of never-smoking adolescents had either smoking parents or peers. About one third was exposed to SHS inside the home and approxidmately 44% were exposed to SHS outside the home. Approximately 90% of never-smoking youth were knowledgeable

Table 2
Susceptibility to smoking among never-smoking youth by WHO region, Global Youth Tobacco Survey, 1999–2008

WHO region	Never-smoking youth		Susceptibility to smoking
	Unweighted count (n)	Weighted count (N)	Weighted percentage (%)
Global	356,414	70,025,268	12.5
Africa	69,649	3,719,593	11.3
Eastern Mediterranean	39,571	6,109,435	14.2
Europe	64,189	2,654,697	29.8
The Americas	83,796	4,418,055	26.5
Southeast Asia	71,080	49,681,806	10.1
Western Pacific	28,129	3,441,682	14

about the harmful effects of smoking and SHS, and 63% were exposed to antismoking media messages. Approximately 79% of youth supported SFPs, and about 58% received antismoking education from their schools (data not shown).

Table 3 presents the bivariate analyses of factors associated with smoking susceptibility among never-smoking youth. Overall, all variables in the bivariate analyses, except support for SFPs ($p = .73$) and age ($p = .07$), were significantly associated with smoking susceptibility among never-smoking youth. Among males, all variables, except support for SFPs ($p = .45$) and school antismoking education ($p = .23$), were significantly associated with susceptibility to smoking. Similarly, among females, all variables, except support for SFPs ($p = .14$), knowledge about the harmful effects of smoking and SHS ($p = .68$), exposure to antismoking media messages ($p = .30$), school antismoking education ($p = .08$), and age ($p = .97$), were significantly associated with susceptibility to smoking.

Table 4 presents the results of multivariable logistic regression analyses. In the total study population, never-smoking males were significantly associated with increased susceptibility to smoking (AOR 1.23, 95% CI 1.11–1.37) than females. Compared with never-smoking adolescents in the Americas, those in the Africa (AOR .46, 95% CI .41–.51), Eastern Mediterranean (AOR .57, 95% CI .51–.63), Southeast Asia (AOR .36, 95% CI .31–.41) and Western Pacific (AOR .44, 95% CI .40–.49) regions were significantly associated with decreased susceptibility to smoking. Never-smoking youth who had either smoking parents or peers were most significantly associated with increased smoking susceptibility (AOR 2.58, 95% CI 2.32–2.87), followed by those who were exposed to SHS inside the home (AOR 1.53, 95% CI 1.42–1.64), those who received tobacco industry promotions (AOR 1.51, 95% CI 1.24–1.84) and those who were exposed to SHS outside home (AOR 1.44, 95% CI 1.29–1.60), respectively. In contrast, never-smoking youth who supported SFPs were significantly associated with decreased susceptibility to smoking (AOR .82, 95% CI .71–.96), followed by those who received antismoking education in schools (AOR .87, 95% CI .77–.98). In addition, we found that never-smoking adolescents who were exposed to antismoking media messages were significantly associated with increased susceptibility to smoking (AOR 1.18, 95% 1.09–1.29) compared with those who were not exposed.

Among males, those who had smoking parents or peers (AOR 2.45, 95% CI 2.19–2.73), those who were exposed to SHS inside (AOR 1.61, 95% CI 1.45–1.79) or outside (AOR 1.42, 95% CI 1.25–1.62) the home, and tobacco industry promotions (AOR 1.58, 95% CI 1.27–1.96) were significantly associated with increased susceptibility to smoking. In addition, those who were exposed to antismoking media messages were associated with increased susceptibility to smoking (AOR 1.20, 95% CI 1.07–1.34). In addition, male never-smoking adolescents aged 13 years were significantly associated with decreased smoking susceptibility compared with 15-year-olds (AOR .81, 95% CI .68–.97). Never-smoking adolescent males who either supported SFPs or received antismoking media messages were not significantly associated with decreased susceptibility to smoking.

Table 3
Bivariate analysis of factors associated with susceptibility to smoking among never-smoking youth aged 13–15 years, Global Youth Tobacco Survey, 1999–2008

Measures	Susceptibility to smoking		
	Total (N = 356,414)	Males (N = 156,401)	Females (N = 200,013)
	% (n)[b]	% (n)	% (n)
Parental or peer smoking	24.2 (33,519)	15.2 (16,501)	9 (17,018)
Exposure to SHS[a] at home	19.7 (29,945)	11.4 (13,889)	8.3 (16,056)
Exposure to SHS[a] outside home	17.3 (40,624)	10 (18,717)	7.3 (21,907)
Support for smoke-free policy	12.5 (52,519)	7.2 (24,321)	5.3 (28,198)
Knowledge about harmful effects of smoking and SHS[a]	12.6 (59,879)	7.2 (27,805)	5.4 (32,074)
Exposure to antismoking media messages	13.2 (39,671)	7.8 (18,990)	5.4 (20,681)
Tobacco industry promotions	18.7 (6,969)	11.8 (3,674)	6.9 (3,295)
School antismoking education	11.9 (38,878)	6.8 (17,607)	5.1 (21,271)
Age			
13 years	11.6 (21,503)	6.1 (9,854)	5.5 (11,649)
14 years	12.6 (24,837)	7.2 (11,524)	5.4 (13,313)
15 years	13.5 (18,413)	8.2 (8,912)	5.3 (9,501)
WHO Region			
Africa	11.3 (7,470)	5.3 (3,701)	6 (3,769)
Eastern Mediterranean	14.2 (5,785)	8.1 (3,144)	6.1 (2,641)
Europe	29.8 (20,232)	11.7 (7,972)	18.1 (12,260)
The Americas	26.5 (20,640)	12.6 (9,169)	13.9 (11,471)
Southeast Asia	10.1 (7,081)	6.4 (4,273)	3.7 (2,808)
Western Pacific	14 (3,545)	7.6 (2,031)	6.4 (1,514)

See Table 1 for details of variable dichotomization (yes/no).
[a] SHS = secondhand smoke.
[b] Weighted percentage, and unweighted frequencies in parentheses.

Table 4
Factors associated with susceptibility to smoking among never-smoking youth, Global Youth Tobacco Survey

	Total	Males	Females
	Adjusted OR 95% CI	Adjusted OR (95% CI)	Adjusted OR (95% CI)
Parental or peer smoking (Ref: nonsmoking parents or peers)	2.58*** (2.32–2.87)	2.45*** (2.19–2.73)	2.72*** (2.29–3.22)
Exposure to SHS inside home (Ref: no exposure to SHS inside home)	1.53*** (1.42–1.64)	1.61*** (1.45–1.79)	1.43*** (1.29–1.58)
Exposure to SHS outside home (Ref: no exposure to SHS outside home)	1.44*** (1.29–1.60)	1.42*** (1.25–1.62)	1.45*** (1.27–1.65)
Support for smoke-free policies (Ref: no support for smoke-free policies)	.82* (.71–.96)	.94 (.76–1.15)	.69** (.57–.84)
Knowledge about the harmful effects of smoking and SHS (Ref: no knowledge about the harmful effects of smoking and SHS)	1.01 (.86–1.19)	1.08 (.90–1.28)	.95 (.70–1.28)
Exposure to antismoking media messages (Ref: no exposure to antismoking media messages)	1.18** (1.09–1.29)	1.20** (1.07–1.34)	1.17* (1–1.37)
Tobacco industry promotions (Ref: no exposure to tobacco industry promotions)	1.51*** (1.24–1.84)	1.58*** (1.27–1.96)	1.42** (1.08–1.86)
School antismoking education (Ref: no antismoking education in schools)	.87* (.77–.98)	.90 (.78–1.04)	.81* (.67–.99)
Age			
13 years	.90 (.76–1.07)	.81* (.68–.97)	1.04 (.76–1.14)
14 years (Ref: 15 years)	.93 (.83–1.04)	.88 (.75–1.03)	1 (.81–1.25)
Males (Ref: females)	1.23** (1.11–1.37)		
WHO Region			
Africa	.46*** (.41–.51)	.49*** (.43–.57)	.43*** (.37–.50)
Eastern Mediterranean	.57** (.51–.63)	.65*** (.56–.75)	.50*** (.41–.60)
Europe	1.05 (.95–1.15)	.89 (.79–1.01)	1.18** (1.06–1.31)
Southeast Asia	.36*** (.31–.41)	.40*** (.34–.47)	.31*** (.27–.37)
Western Pacific (Ref: The Americas)	.44*** (.40–.49)	.54*** (.47–.63)	.37*** (.32–.43)

CI = Confidence Intervals; OR = Odds Ratio; SHS = secondhand smoke.
* $p < .05$.
** $p < .01$.
*** $p < .001$.

Among females, those who had smoking parents or peers (AOR 2.72, 95% CI 2.29–3.22), those who were exposed to SHS inside (AOR 1.43, 95% CI 1.29–1.58) and outside (AOR 1.45, 95% 1.27–1.65) the home, and tobacco industry promotions (AOR 1.42, 95% CI 1.08–1.86) were significantly associated with increased susceptibility to smoking. In contrast, those who supported SFPs (AOR .69, 95% CI .57–.84) and those who received school antismoking education (AOR .81, 95% CI .67–.99) were significantly associated with decreased susceptibility to smoking. Unlike results in the overall study population and adolescent males, females in the European region were significantly associated with increased smoking susceptibility compared with those in the Americas (AOR 1.18, 95% CI 1.06–1.31).

Discussion

Tobacco use is a growing epidemic with more than one billion smokers worldwide, most of whom reside in the LMICs [3]. Up to 90% of smokers begin smoking as adolescents [2], and it is important to identify the causes of this phenomenon. Previous studies have identified several factors that influence youth transition from smoking experimentation to a regular smoking, including susceptibility to smoking [10,27]. While studies on youth susceptibility to smoking are mostly confined to high-income countries [10,28], there are only a few such studies for LMICs [4,6]. In this study, we estimated the susceptibility to smoking among never-smoking youth worldwide and also identified factors associated with such behavior. We found that 12.5% (8.8 million) of never-smoking youth worldwide aged 13–15 years were susceptible to smoking. Regardless of gender, while youth who were exposed to parental or peer smoking, exposure to SHS both inside or outside home, and to tobacco industry promotions were significantly associated with increased susceptibility to smoking, those who supported SFPs, received antismoking education in schools, and lived outside the Americas were significantly associated with decreased susceptible to smoking. With globalization of tobacco and increased tobacco industry activities in LMICs, the study findings have the potential to inform early intervention efforts and prevent never-smoking youth from initiating smoking and transitioning to regular smoking. The global-level nature of the study suggests that the findings will be useful in directing development of new policies and tobacco control programs targeted at youth in pre-experimentation stages of smoking learned behavior.

Gender-wise analyses in this study illuminated similarities and differences in the factors associated with susceptibility to smoking among never-smoking adolescent males and females. Consistent with previous studies, we found males were more susceptible to smoking than females, a finding which supports focusing special attention on gender while developing interventions [29]. As shown by previous studies, parental or peer smoking behavior was significantly associated with increased susceptibility to smoking among males and females [6,14,30,31]. Although causal relationship could not be established, these findings suggest that it is possible that the smoking behavior of family members and/or friends creates a perception for adolescents that smoking is socially acceptable, and makes tobacco products more accessible to them, which might encourage them to smoke their first cigarette. In addition, previous studies identified peer smoking as more influential in smoking initiation and regular smoking than parental smoking [32,33]. Therefore, future studies are warranted in this direction because this study did not delineate associations between never-smoking youth susceptibility to smoking, and parental or peer smoking behavior separately. Nevertheless, the findings suggest the need for parental and peer education to help them either quit smoking entirely or at least abstain from smoking in front of their children or peers. Moreover, the public health community should promote tobacco control

programs that empower never-smoking youth to resist peer pressures, that prevent youth access to tobacco products, and that facilitate protective parenting practices.

We found that tobacco industry promotions were associated with increased smoking susceptibility among never-smoking males and females, which is similar to earlier studies [14,28,34]. This finding supports the 2012 World No Tobacco Day theme—"*Tobacco industry interference*", which is aimed at exposing and countering the tobacco industry's attempts to undermine tobacco control strategies. This finding demands attention because earlier studies had identified a dose-response relationship between an increased smoking susceptibility index and increased tobacco use among never-smokers when adolescents were more exposed to tobacco promotion campaigns [35]. The tobacco industry promotion efforts can be prevented by developing and implementing comprehensive tobacco bans in all countries along with stronger regulation of tobacco company practices in accordance with the Articles 9, 10, 11, and 13 of the WHO Framework Convention on Tobacco Control (FCTC), the first international public health treaty negotiated under the auspices of the WHO [36].

Exposure to SHS inside or outside the home was associated with increased susceptibility to smoking among never-smoking males and females, respectively. This finding is consistent with earlier studies that identified increased cigarette smoking initiation among never-smoking subjects when exposed to SHS [37]. SHS contains over 4,000 chemicals, of which 70 are known carcinogens and poses serious health hazards to children, yet only 1 in 30 adolescents worldwide live in countries protected by comprehensive SFPs [38]. Therefore the study findings support the need to create smoke-free environments worldwide, including a ban on smoking in vehicles with youth. This is because, while SFPs protect adolescent never-smokers from exposure to SHS, the study findings suggest that they have the potential to decrease youth susceptibility to smoking and prevent transition to future regular smokers. In the study, we also found that never-smoking adolescents who were exposed to antismoking media messages were associated with increased susceptibility to smoking. Although this finding is consistent with earlier studies [6,31], it provides important evidence about the delivery and receptivity of current antismoking messages. This positive association potentially might be due to interference of tobacco companies in the delivery of antismoking messages and nullifying their effect on adolescents' attitudes and behavior toward smoking [22]. Because this positive association might vary across different WHO regions or countries, future studies are needed to evaluate this finding. Along with this, it is also important that antismoking messages be appropriately delivered through media without interference by tobacco companies.

In the study, we identified never-smoking youth who supported SFPs or received school antismoking education as significantly associated with decreased susceptibility to smoking. Although support for SFPs was inversely associated with susceptibility to smoking among never-smoking youth, the relationship was significant only for females. This finding is similar to that in a previous study conducted to understand worldwide youth attitudes toward SFPs among smokers, where susceptibility to smoking was found to be associated with decreased likelihood of support for SFPs [18]. This implies that, in addition to protecting nonsmokers and assisting smokers to quit smoking, SFPs have the capacity to reduce the propensity of never-smoking youth to initiate smoking. With respect to school antismoking education, we found an inverse association with susceptibility to smoking (significant only among females). Although previous studies provided mixed results on the effectiveness of school antismoking programs on smoking susceptibility [39], this finding reinforces the importance of antismoking education in educational institutions in curbing the rising tobacco epidemic worldwide. In addition, we found that never-smoking youth in the Africa, Europe (males only), Eastern Mediterranean, Southeast Asian, and Western Pacific regions were associated with decreased susceptibility from those in the Americas, which might be attributed to cultural variability of tobacco use. These findings support efforts to prevent tobacco industry penetrations into LMICs where cigarette smoking is a relatively recent phenomenon, and to identify and reinforce cultural factors that are protective against youth smoking.

Although not significant in this study, the positive association between knowledge about the harmful effects of smoking and SHS merits attention. Previous studies found knowledge about the harmful effects of smoking provides protection against smoking initiation and increases the propensity to quit smoking [4,40] and support for SFP [18]. In particular, in a study similar to ours Koh et al. identified that knowledge about the harmful effects of smoking increased support for SFPs among youth worldwide [18]. The difference in findings might be due to differences in study populations, that is, all youth were used in Koh et al. study, and never-smoking youth in the current study. Therefore, along with ongoing research efforts to study the role of knowledge about smoking harm among youth, it is important to evaluate its role among never-smoking youth, given that imparting knowledge about harmful effects of tobacco use has been one of the key tobacco control strategies.

The study has strengths and limitations that merit discussion. It is the first study to estimate and identify factors associated with smoking susceptibility behavior among never-smoking youth on a global scale. However, the GYTS is a school-based survey and presents information only about youth attending schools. Those who did not attend schools or those attending schools but absent on the day the survey was conducted were not captured. In addition the students' responses are self-reported and subject to recall bias. Moreover, 17 questions in the GYTS 'core' questionnaire were pooled into 11 measures (Table 1) based on the existing literature, which limits identifying differences in susceptibility when exposed to individual components. The study used measures at individual-level to understand susceptibility to smoking among never-smoking youth, along with country-level (ecological) measures such as development status and cultural variability across countries; presence or absence of national tobacco control plans of respective countries were not taken into account. Future studies are warranted in this direction. Furthermore, the GYTS is a cross-sectional study and, therefore, cannot be used to make firm inferences regarding causality. Nevertheless, the study sheds light on smoking susceptibility behavior among never-smoking youth at a global level that is useful for the public health community and policymakers worldwide.

The study identified that 12.5% (8.8 million) of never-smoking youth aged 13–15 years were susceptible to initiate smoking. Susceptibility to smoking among youth varied across the WHO regions with least and highest rates in the Southeast Asia and the European regions, respectively. Parental or peer smoking behavior, tobacco industry promotions, and exposure to SHS inside or outside the home were significantly associated with

increased smoking susceptibility, while antismoking school education and support for SFPs were associated with decreased susceptibility. With sparse research on susceptibility to smoking in LMICs and globally, the GYTS data were used to identify these factors, thereby addressing a major gap in the scientific literature. The study findings provide evidence regarding the importance of understanding the early stages of cigarette smoking learned behavior, and of implementing policies and programs that limit youth exposure to SHS, or modifying social norms regarding tobacco use to make cigarette smoking socially unacceptable. Hence, developing well-executed antismoking campaigns, comprehensive SFPs, comprehensive bans on tobacco industry advertising and promotion strategies, together with antismoking education in the schools in accordance with the FCTC, are some of the important public health efforts that may prevent never-smoking youth worldwide from becoming susceptible to smoking.

Acknowledgments

S.P.V. conducted this study as a part of his dissertation work [41] for Doctor of Public Health in Epidemiology at East Tennessee State University. S.P.V. would like to thank Drs. Randy Wykoff, Robert Pack, and Amal Khoury for their support; and East Tennessee State University for providing the logistical support for conducting the study. In addition, the authors would like to thank all personnel, including study participants and organizations involved with the Global Youth Tobacco Survey. In particular, the authors would like to acknowledge the U. S. Centers for Disease Control and Prevention for making the data publicly available through the GTSSdata portal.

Supplementary Data

Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.jadohealth.2013.07.036.

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